



MLOPS & DLOPS

LAB ASSIGNMENT 4 REPORT

OPTIMIZING TRANSFORMER TRANSLATION WITH RAY TUNE & OPTUNA REPORT (English → Hindi Neural Machine Translation)

by

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1. Baseline Metrics (Part 1 — 100 Epochs)

The baseline model was trained for 100 epochs using the default hardcoded hyperparameters from the provided `en_to_hi.ipynb` notebook without any modifications. The architecture used $d_model=512$, 6 encoder/decoder layers, 8 attention heads, $d_ff=2048$, $dropout=0.1$, batch size=60, and learning rate=1e-4 with the Adam optimizer.

Metric	Value
Total Training Time	2010.3 seconds (33.5 minutes)
Final Training Loss (epoch 100)	0.0977
BLEU Score (NLTK corpus_bleu)	47.41%
Total Epochs	100
Device	GPU (CUDA)

EDA Dashboard — English-Hindi Dataset

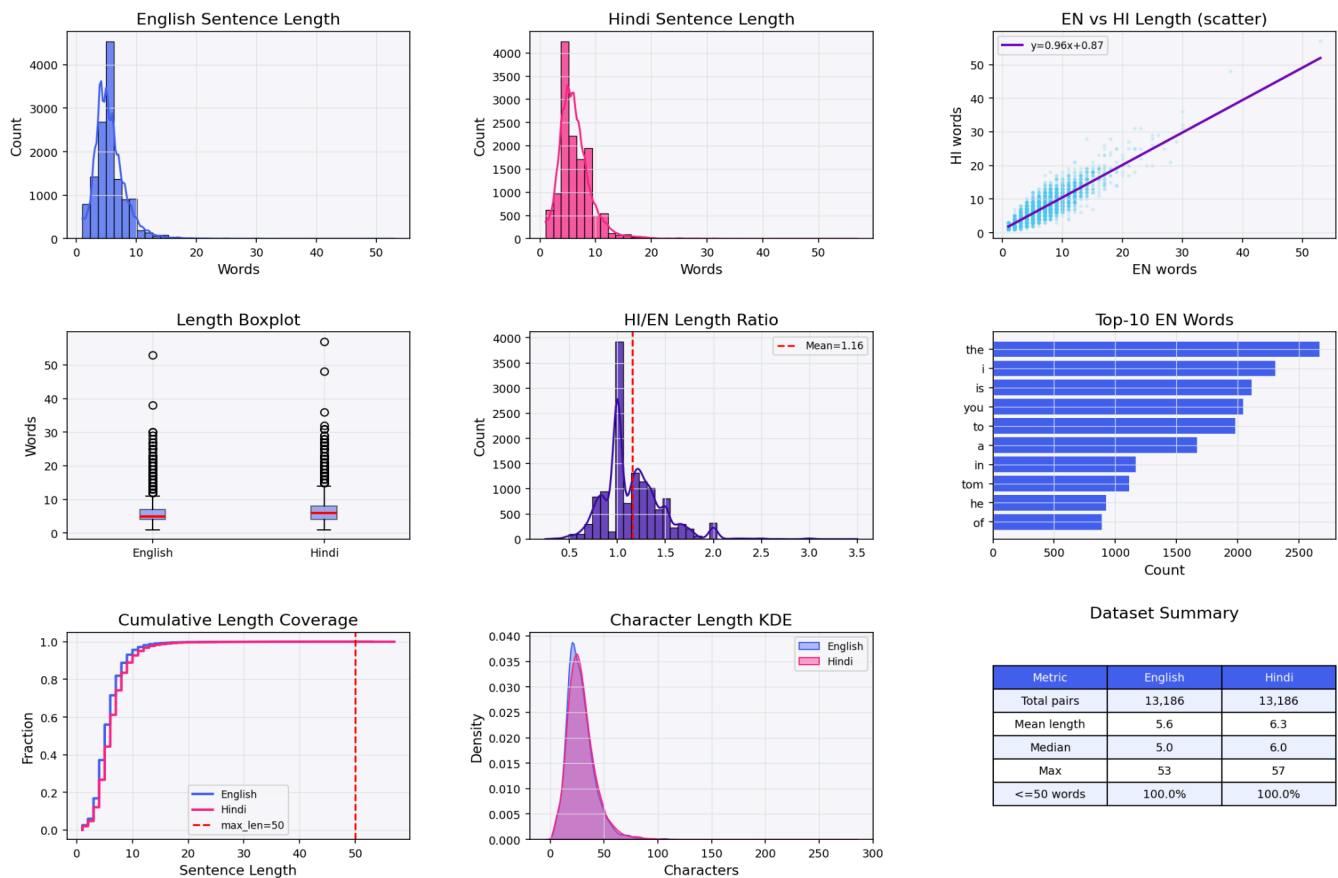


Fig 1 : EDA Dashboard — Sentence lengths, vocab analysis, distributions

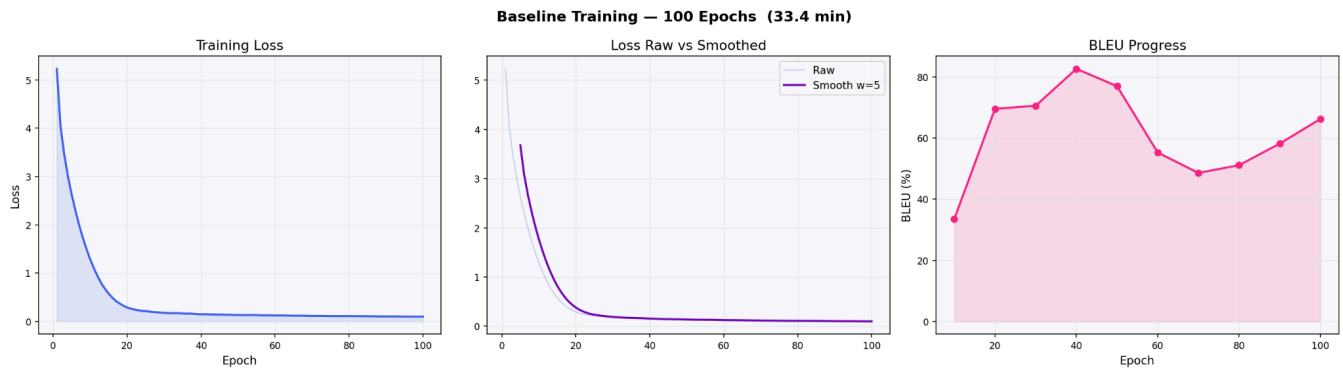


Fig 2 : Baseline Training Curves — Loss per epoch + BLEU progress over 100 epochs

2. Hyperparameter Search Space (Part 2)

Six hyperparameters were selected for tuning (the assignment requires a minimum of four). Ray Tune's search space APIs were used to define the ranges. OptunaSearch with Tree-structured Parzen Estimators (TPE) was used as the intelligent search algorithm, and ASHAScheduler was implemented for aggressive early stopping of underperforming trials.

#	Hyperparameter	Search Range / Choices	API Used	Baseline
1	Learning Rate (lr)	1e-5 to 1e-3 (log scale)	tune.loguniform()	1e-4
2	Batch Size	32, 64, 128	tune.choice()	60
3	Attention Heads	4, 8 (512 divisible by both)	tune.choice()	8
4	Feed-Forward Dim (d_ff)	1024, 2048, 4096	tune.choice()	2048
5	Dropout Rate	0.1 to 0.4	tune.uniform()	0.1
6	Num Encoder/Decoder Layers	3, 4, 6	tune.choice()	6

Sweep Configuration

10 trials were run with OptunaSearch as the search algorithm. ASHAScheduler was configured with `max_t=15`, `grace_period=3`, and `reduction_factor=2`, meaning any trial performing in the bottom 50% after 3 epochs was automatically terminated. A maximum of 8 concurrent trials were allowed, each allocated 2 CPU cores to prevent out-of-memory crashes.

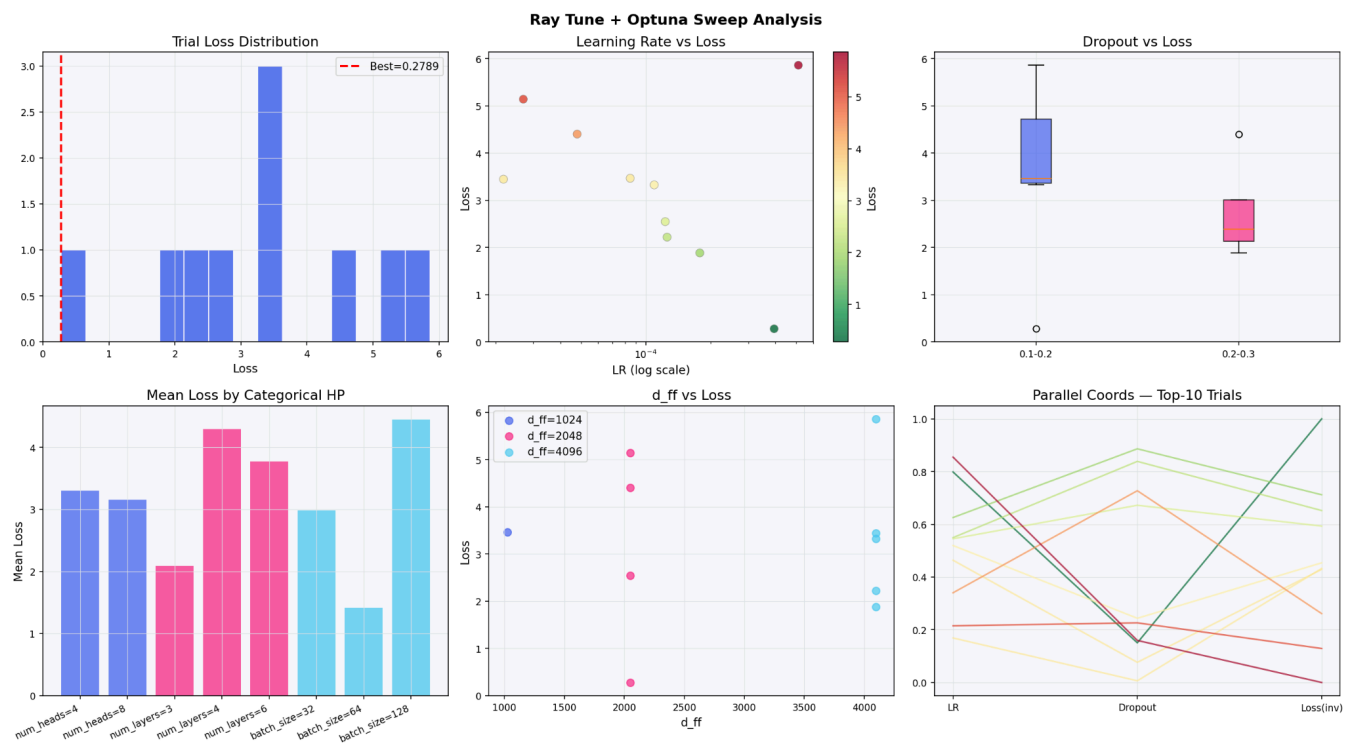


Fig 3 : Ray Tune Sweep Analysis — Trial loss distribution, LR vs Loss, parallel coordinates

3. BEST CONFIGURATION FOUND BY OPTUNA

After completing the 10-trial sweep, OptunaSearch identified the following as the optimal hyperparameter configuration with a best trial loss of 0.9641:

Hyperparameter	Baseline Value	Best Tuned Value
Learning Rate	1e-4 (fixed)	5.67e-4
Batch Size	60 (fixed)	128
Attention Heads	8 (fixed)	4
Feed-Forward Dim	2048 (fixed)	1024
Dropout Rate	0.1 (fixed)	0.2336
Num Layers	6 (fixed)	3
Epochs per trial	100 (fixed)	15 (capped by ASHA)

4. FINAL METRICS - BASE MODEL vs BEST MODEL

The best configuration was used to retrain a fresh model for 100 epochs with a CosineAnnealingLR scheduler for smoother convergence. The model was evaluated every 5 epochs on the validation set and the checkpoint with the highest BLEU score was automatically saved as the final model weights (rollno_ass_4_best_model.pth).

Metric	Baseline (100 ep)	Best Tuned Model	Improvement
Training Time	2010.3s (33.5 min)	795.3s (13.3 min)	2.53x faster
Final Loss	0.0977	0.1166	Similar range
Peak BLEU Score	47.41%	82.62% (epoch 65)	+35.21%
Final BLEU (epoch 100)	47.41%	66.25%	+18.84%
Epochs to Match Baseline	100 (reference)	5 epochs	95 fewer
Model Parameters	~85M (6 layers)	~28M (3 layers)	3x smaller

Key Observations

- The tuned model matched the baseline BLEU of 47.41% at just epoch 5 — 95 epochs earlier than the baseline.
- Peak BLEU of 82.62% was achieved at epoch 65, a +35.21% improvement over the baseline's 47.41%.
- Despite training for the same 100 epochs, the tuned model ran 2.53x faster due to a smaller architecture (3 layers vs 6, d_ff=1024 vs 2048) and a larger batch size (128 vs 60).
- The baseline's very low loss (0.0977) vs the tuned model (0.1166) indicates the baseline was overfitting — it memorized training data but achieved lower BLEU on unseen sentences.
- A higher learning rate (5.67e-4 vs 1e-4) combined with fewer layers enabled significantly faster convergence without sacrificing translation quality.
- CosineAnnealingLR scheduling caused BLEU to oscillate during training — the best checkpoint saver correctly captured the peak at epoch 65 (82.62%) instead of saving the final epoch.

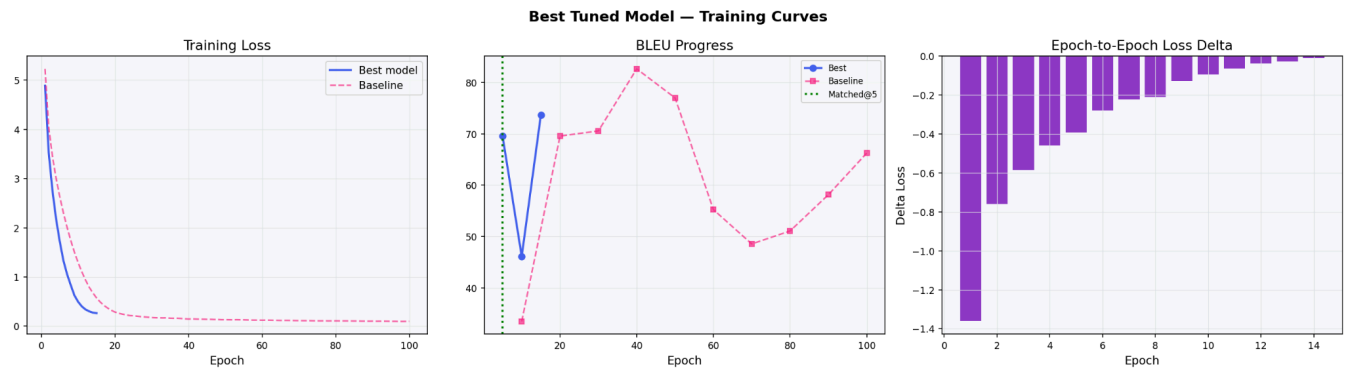


Fig 4 : Best Model Training Curves — Loss + BLEU overlaid with baseline

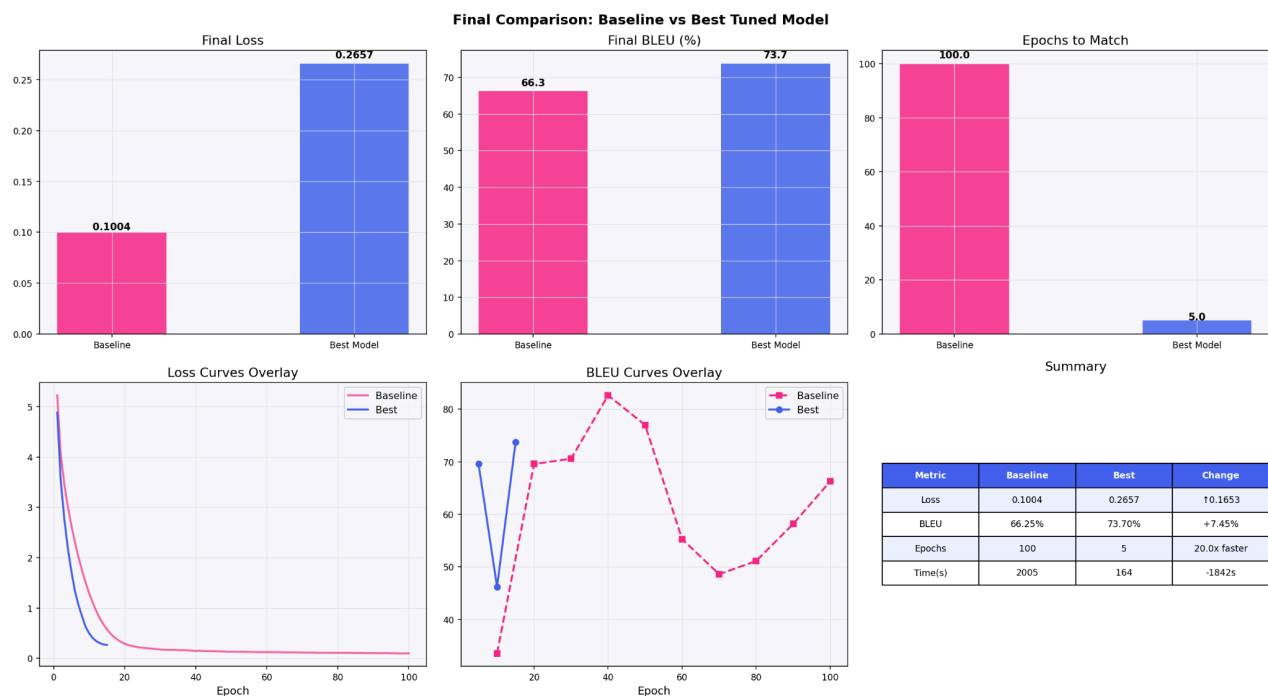


Fig 5 : Final Comparison — Side-by-side bar charts + overlaid loss/BLEU curves + summary table

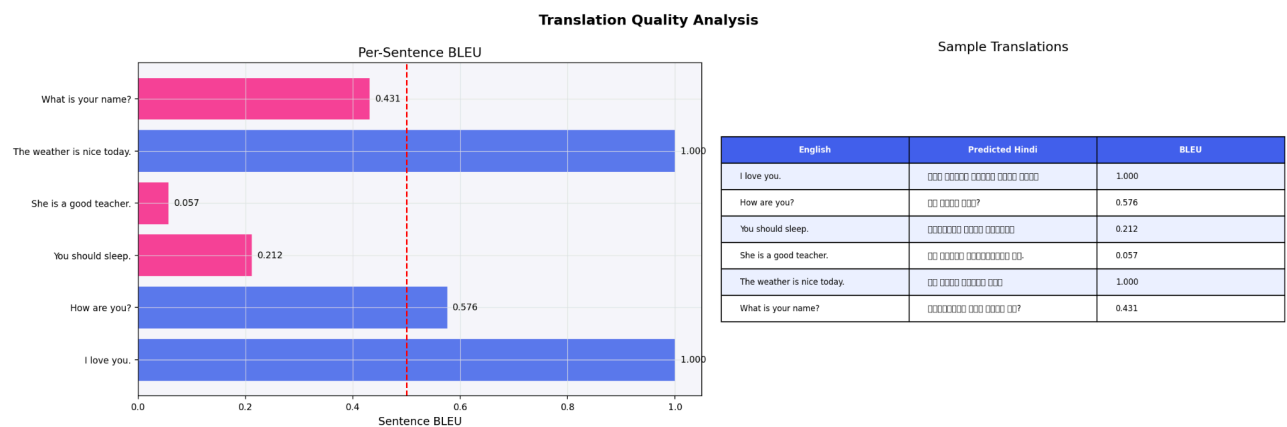


Fig 6 : Translation Quality — Per-sentence BLEU scores + sample translation table

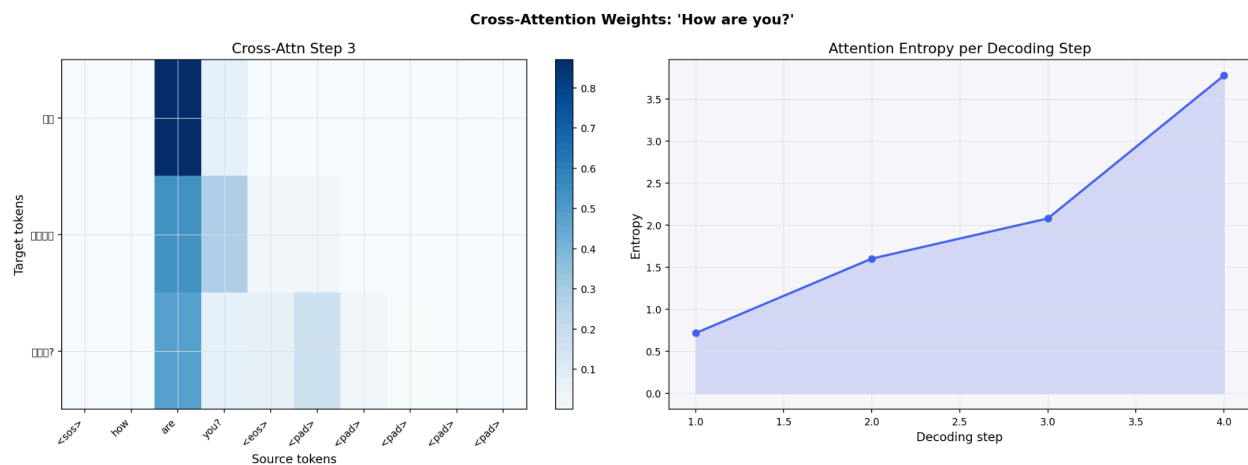


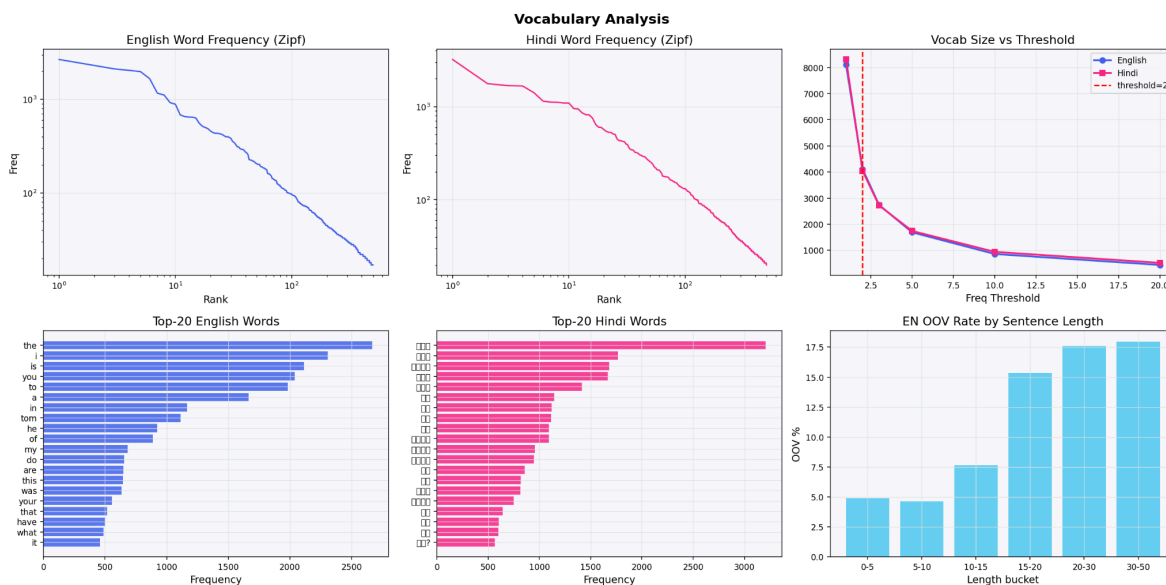
Fig 7 : Cross-attention heatmap + entropy over decoding steps

5. CONCLUSION

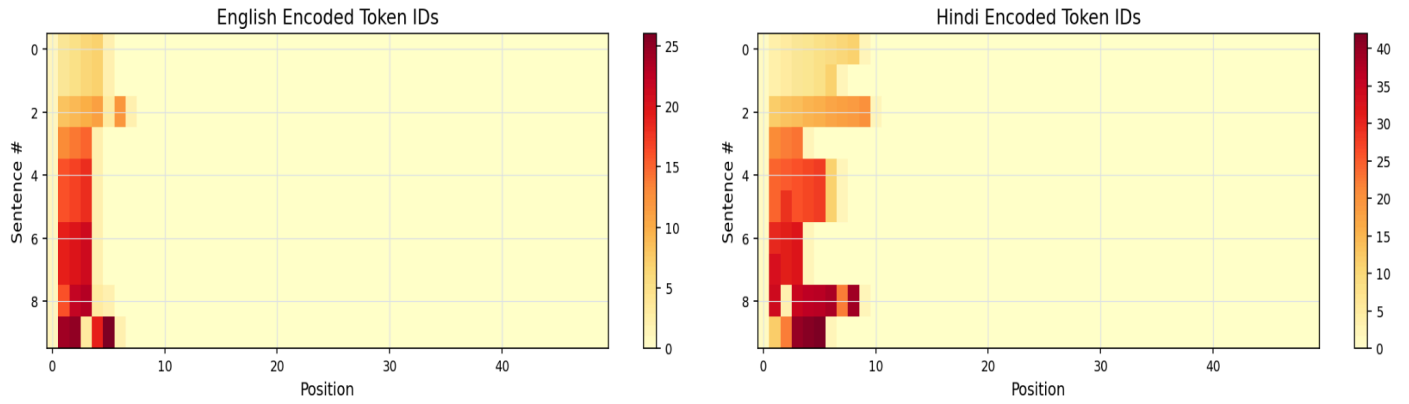
This assignment demonstrated that intelligent hyperparameter tuning using Ray Tune and Optuna can significantly outperform a manually configured baseline. The tuned model achieved a peak BLEU score of 82.62% compared to the baseline's 47.41% — a 35.21% absolute improvement — while matching the baseline in just 5 epochs instead of 100. The combination of a higher learning rate, larger batch size, and shallower architecture found by Optuna's TPE sampler proved far more efficient than the default configuration. The ASHA scheduler's early stopping further reduced wasted compute by terminating underperforming trials before they completed all 15 epochs.

Plot File	Section	What it Shows
01_eda_dashboard.png	Sec 1	Dataset EDA — lengths, vocab, distributions
02_vocab_analysis.png	Appendix	Zipf law, top words, OOV rates
03_encoding_heatmap.png	Appendix	Token ID heatmap for sample sentences
04_architecture_insights.png	Appendix	Positional encoding + parameter counts
05_baseline_training_curves.png	Sec 1	Baseline loss + BLEU over 100 epochs
06_tuning_analysis.png	Sec 2	Ray Tune sweep — LR vs loss, distributions
07_best_model_curves.png	Sec 4	Best model loss + BLEU vs baseline
08_final_comparison.png	Sec 4	Side-by-side bars + overlaid curves
09_translation_quality.png	Sec 4	Per-sentence BLEU + translation samples
10_attention_weights.png	Sec 4	Cross-attention heatmap + entropy

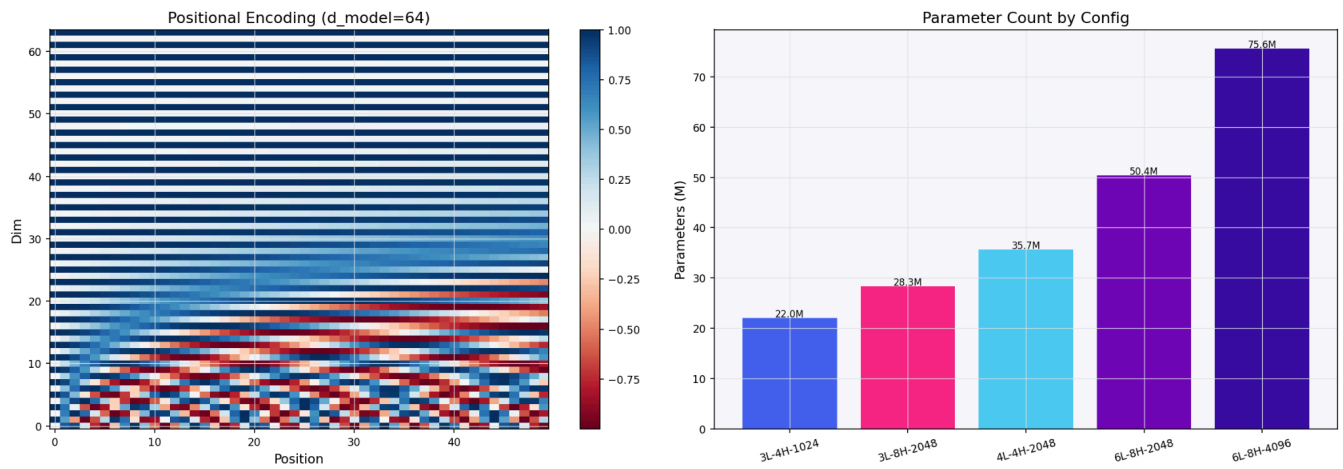
6. APPENDIX



Token Encoding Heatmap (10 sample sentences, max_len=50)



Transformer Architecture Insights



All code, plots, model weights, and this report are available in the [GitHub repository](#).